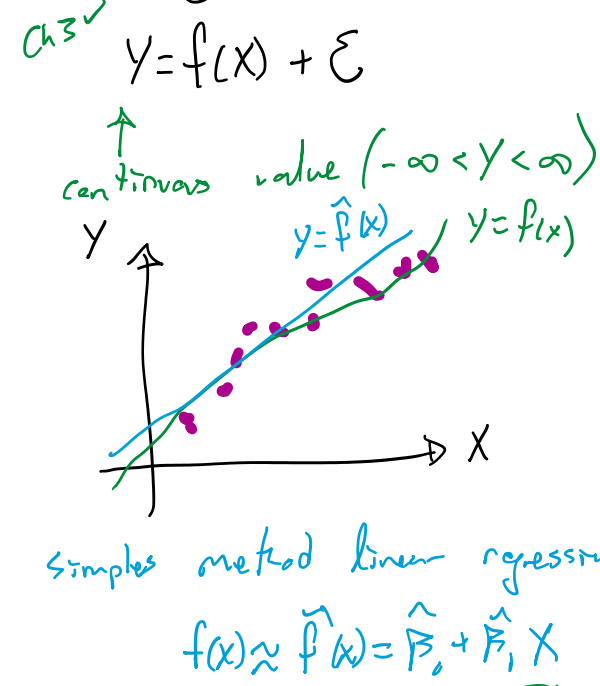


Wk4.1 Logistic regression

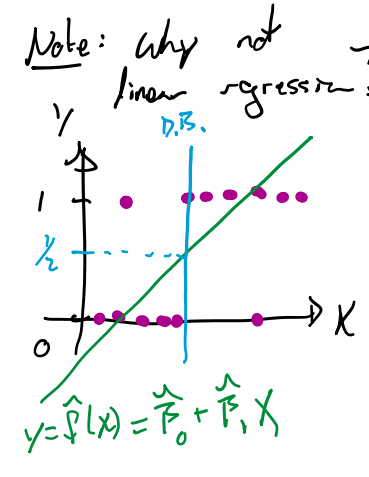
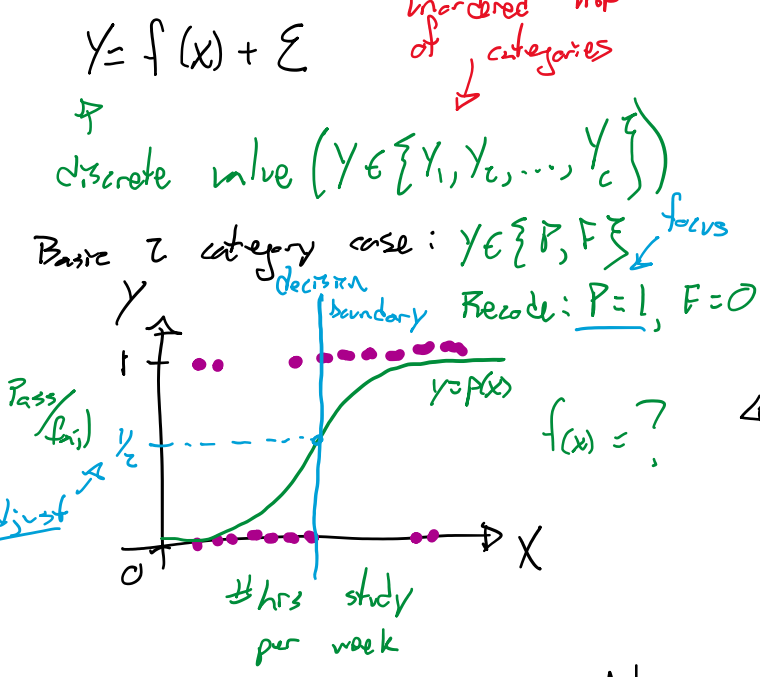
Tuesday, September 29, 2026 9:20 AM

2 groups of supervised learning tasks:

① Regression



② Classification



Ch4 Classification

3 main approaches:

- ① Model $P(Y=1 | X)$ (logistic regr.) directly
- ② Model $P(Y=1 | X)$ indirectly via $P(X | Y=1)$ (linear/binary discriminant analysis or Naive Bayes)
- ③ KNN non-parametric. Assume indep.

Intro $\rightarrow$ Bayes Theorem

Logistic regression:

Goal is to model

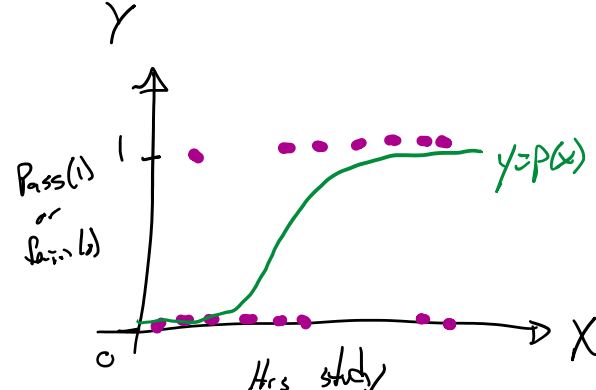
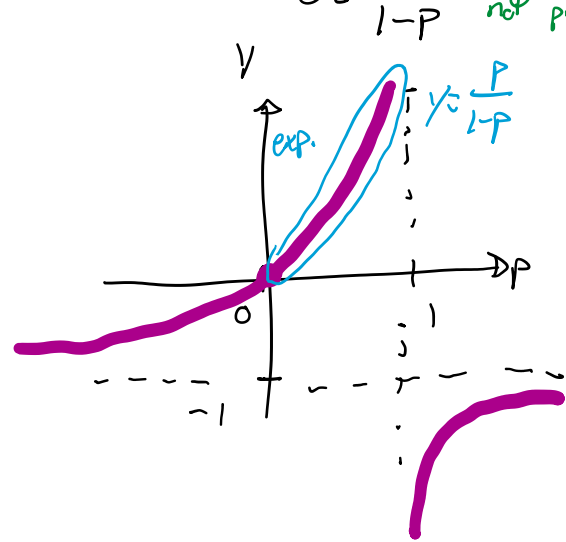
$$p = p(x) = P(Y=1 | X)$$

via ① Data transformations

② Linear regression

Data trans: ① Prob $0 \leq p \leq 1$

② Odds ratio $0 \leq \frac{p}{1-p} < \infty$



Ex: Roll 6, win

$$p_0 = 1/6$$

$$\frac{p_0}{1-p_0} = \frac{1/6}{1-1/6} = \frac{1/6}{5/6} = 1/5$$

odds 1:5

loss win 5(+1)

③ Log-odds: $\ln(\frac{p}{1-p})$ is linear-ish

Logistic regression assumes $\ln(\frac{p}{1-p}) = \beta_0 + \beta_1 X$

Solve for p:

$$\ln(\frac{p}{1-p}) = \beta_0 + \beta_1 X$$

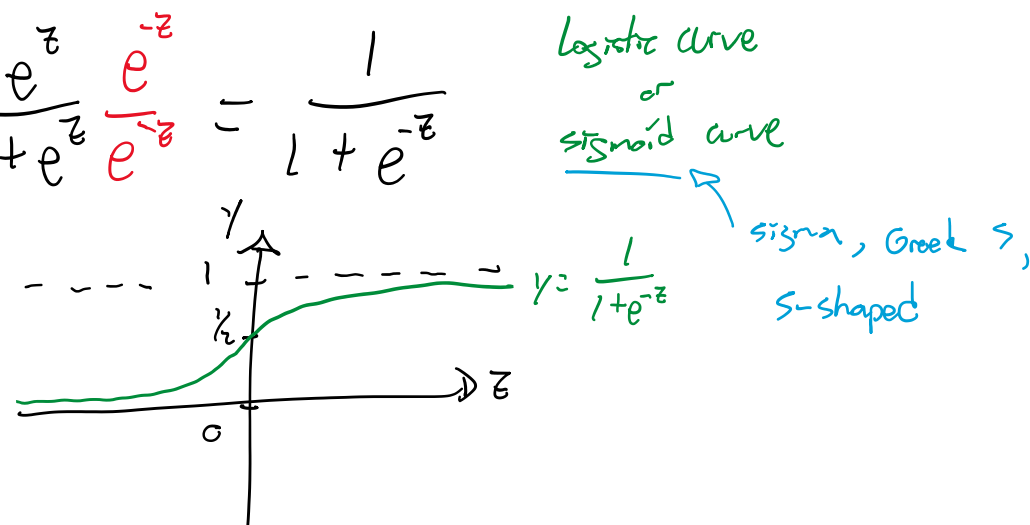
$$\frac{p}{1-p} = e^{\beta_0 + \beta_1 X}$$

$$p = (1-p)e^{\beta_0 + \beta_1 X}$$

$$p + p e^{\beta_0 + \beta_1 X} = e^{\beta_0 + \beta_1 X}$$

$$p(1 + e^{\beta_0 + \beta_1 X}) = e^{\beta_0 + \beta_1 X}$$

$$p = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$



So $\ln(\frac{p}{1-p}) = \beta_0 + \beta_1 X \rightarrow p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$

β_i interpretations: β_1 : If increases X one unit, $\ln(\frac{p}{1-p})$ (log odds) increase by β_1

or gives p increase

β_0 : Gives $p(x) = 1/2$ threshold

How to fit to data to get $\hat{\beta}_0, \hat{\beta}_1$?

Because dataset has no $p(x)$, cannot minimize RSS or use least squares. Instead maximize likelihood function.

(How likely data (X_i, Y_i) to true β_0, β_1)

$$l(\beta_0, \beta_1) = \prod_{i=1}^n P(Y_i | X_i, \beta_0, \beta_1)$$

max over possible

$$= P(Y_1 | X_1) \cdot P(Y_2 | X_2) \cdot \dots \cdot P(Y_n | X_n)$$

data indep

$$= \prod_{Y_i=1} P(X_i) \cdot \prod_{Y_i=0} (1 - P(X_i))$$

where $P(X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$

Instead of maximizing $l(\beta_0, \beta_1)$ directly (many product rules), use log transf.

Max log-odds + set $\hat{\beta}_i$'s.

$$\ln(l(\beta_0, \beta_1)) = \sum_{i=1}^n \ln(P(X_i)) + \sum_{i=1}^n \ln(1 - P(X_i))$$

n probs.

$$= \sum_{i=1}^n (Y_i \ln(P(X_i)) + (1 - Y_i) \ln(1 - P(X_i)))$$

trade

In end, fitted model $p(x) = \frac{1}{1 + e^{-(\hat{\beta}_0 + \hat{\beta}_1 X)}}$ via

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\max_{(\beta_0, \beta_1)} \ln(l(\beta_0, \beta_1))$$

by numerical methods.

2 generalizations ① P-predictors $X = (X_1, \dots, X_p)$

② $C > 2$ classes/categories

↳ multinomial logistic regression

Multiple logistic regression

$$\ln(\frac{p}{1-p}) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = \vec{\beta}^T \vec{X}$$

$$\vec{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$$

$$\vec{X} = \begin{bmatrix} 1 \\ X_1 \\ \vdots \\ X_p \end{bmatrix}$$

$$p(x) = \frac{1}{1 + e^{-\vec{\beta}^T \vec{X}}}$$

Still fit $\vec{\beta}$ via max log likelihood.

Multinomial logistic regression $C > 2$ classes. $Y \in \{Y_1, Y_2, \dots, Y_C\}$

Recall: $C=2$, $p(x) = P(Y=1 | X)$ & $P(Y=0 | X) = 1 - p(x)$

For $C > 2$, let class Y_C be the reference class & $\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$

β_i 's smth interp as $C=2$.

$$1 \leq i \leq C-1 \quad p_i(x) = P(Y=Y_i | X) = \frac{e^{\beta_{i0} + \beta_{i1} X_1 + \dots + \beta_{ip} X_p}}{1 + e^{\beta_{i0} + \beta_{i1} X_1 + \dots + \beta_{ip} X_p}}$$

or $p_C(x) = 1 - \sum_{i=1}^{C-1} p_i(x)$ (probs sum to 1)

A related max softmax approach: No reference class.

$$p_i(x) = P(Y=Y_i | X) = \frac{e^{\beta_{i0} + \beta_{i1} X_1 + \dots + \beta_{ip} X_p}}{\sum_j e^{\beta_{j0} + \beta_{j1} X_1 + \dots + \beta_{jp} X_p}}$$

β_i 's smth interp.